Abhishek Patil
08
124142 43

Page No.
Date

* Home Assignment :- 3 *

Q.A] Attempt following.

→ 4) Let $P(t)$ be vector space of polynomials & for $f(t), g(t)$ in $P(t)$ with inner product $\langle f(t), g(t) \rangle = \int_{-1}^{1} f(t) \cdot g(t)\, dt$ then for $f(t) = t^3$ & $g(t) = t^{-1}$ find $\langle f(t), g(t) \rangle$ & $\|g\|$

→

$f(t) = t^3, \quad g(t) = t$

Inner product ⇒

$$\langle f, g \rangle = \int_{-1}^{1} t^3 \cdot t\, dt = \int_{-1}^{1} t^4\, dt$$

$$= \left[\frac{t^5}{5}\right]_{-1}^{1} = \frac{1}{5} - \left(\frac{-1}{5}\right) = \frac{2}{5}$$

$\|g\|$ ⇒

$$\sqrt{\langle g, g \rangle} \Rightarrow \sqrt{\int_{-1}^{1} t^2\, dt} \Rightarrow \sqrt{\left[\frac{t^3}{3}\right]_{-1}^{1}}$$

$$\Rightarrow \sqrt{\frac{1}{3} - \left(-\frac{1}{3}\right)}$$

$$\Rightarrow \sqrt{\frac{2}{3}}$$

QB] find k for orthogonal.

→ 2) $u = (2, 2, 2)$ & $v = (2, k, 2)$

$u \cdot v = 0$

$\therefore \; 2 \cdot 2 + 2k + 2 \cdot 2 = 0$

$8 + 2k = 0$

$\therefore \boxed{k = -4}$

QC] Apply Gram - Schmidt orthogonalizat^n to find orthogonal basis.

→ 4)

$S = \{ v_1 = t, \; v_2 = 1, \; v_3 = t^2 \}$ $\quad \langle u, v \rangle = \int_{-1}^{1} uv\,dt$

$u_1 = v_1 = t$

$u_2 = v_2 - proj_{u_1}(v_2)$

$u_3 = v_3 - proj_{u_1}(v_3) - proj_{u_2}(v_3)$

Computing, $u_2 = 1 - \dfrac{\langle 1, t \rangle}{\langle t, t \rangle} t$

$\langle 1, t \rangle = \int_{-1}^{1} t\,dt = 0$

$\langle t, t \rangle = \int_{-1}^{1} t^2\,dt = \dfrac{2}{3}$

$\therefore \; u_2 = 1 - 0 \cdot t$

$= 1.$

Computing, $u_3 = t^2 - \dfrac{\langle t^2, t \rangle}{\langle t, t \rangle} t - \dfrac{\langle t^2, 1 \rangle}{\langle 1, 1 \rangle} \cdot 1$

$\langle t^2, t \rangle = \int_{-1}^{1} t^3\,dt = 0$

$\langle t^2, 1 \rangle = \int_{-1}^{1} t^2\,dt = 2/3$

$\langle 1, 1 \rangle = \int_{-1}^{1} 1\,dt = 2$

$\therefore$ ~~$u_3 = t^2 - 0 - \frac{2}{3}$~~ 2

$$u_3 = t^2 - 0 - \frac{2}{3} / 2 \times 1$$

$$\Rightarrow t^2 - \frac{1}{3}$$

$$\therefore \left\{ u_1 = t, \quad u_2 = 1, \quad u_3 = t^2 - \frac{1}{3} \right\}$$

dD] Attempt the following.

3) Find least square sol^{n}s of $AX = B$ where

$$A = \begin{bmatrix} 1 & 1 \\ -1 & 2 \\ 1 & 3 \end{bmatrix} \quad \& \quad B = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ -1 & 2 \\ 1 & 3 \end{bmatrix} \Rightarrow \begin{bmatrix} 3 & 4 \\ 4 & 14 \end{bmatrix}$$

$$A^T B = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 \\ 8 \end{bmatrix}$$

$$A^T A X = A^T B$$

$$\begin{bmatrix} 3 & 4 \\ 4 & 14 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \end{bmatrix}$$

$\therefore \; 3x_1 + 4x_2 = 1 \quad — 1$

$\Rightarrow$ multiply by 14, $\Rightarrow \; 42x_1 + 56x_2 = 14 \quad — 2$

$\therefore \; 4x_1 + 14x_2 = 8 \quad — 3$

$\Rightarrow$ Multiply by $4 \Rightarrow \; 16x_1 + 56x_2 = 32 \quad — 4$

subtract 2 & 4.

$(42x_1 + 56x_2) - (16x_1 + 56x_2) = 14 - 32$

$\Rightarrow \quad 26x_1 = -18$

$\Rightarrow \quad \boxed{x_1 = -\frac{9}{13}}$ //

$\therefore \quad \boxed{x_2 = \frac{10}{13}}$ //

$\therefore$ least square soluth $\Rightarrow X = \begin{bmatrix} -9/13 \\ 10/13 \end{bmatrix}$